



Vilnius University

Faculty of mathematics and informatics

AIS SHIP TRACKING

Collision Analysis

Simonas Arliukas

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1 Data

I import AIS tracking data for the Baltic and North seas. In essence, it is a dataset that tracks vessel states over time. More notably, it checks MMSI which is an identifier provided to each ship, speed over ground, course over ground and heading. All of these variables are used to detect collisions withing the data.

2 Detection strategy

To detect the collision I narrow the dataset to December 2021. In total I end up with around 300 million observation. I further filter by geospatial location. I draw a circle around the center lat: 55.22 and lon: 14.24 with a 50 nautical mile radius. This step specifically filters the vessel in that area. Furthermore, I use time bucketing. Time bucketing is needed because pings might be sent not at the exact second that the ships collide so I check wherever the ships are very close to each other in one minute intervals.

Now once everything is set up I use pyspark to filter the dataset in parallel. Firstly, I check wherever the ship has a speed over ground of more than 0 and speed is less than 50. This is done to filter out vessels that are stationary and ones that are moving at a possible speed. Then I check wherever the combined speed of the ships is more than 14 knots. This is done to spot whenever one ship rams into another ship meaning that at least one ships is moving at a high speed. Furthermore, I check if the heading and COG are more than 36 degrees or less than 324 degrees to filter out ships that are moving in parallel and staying close to each other. A police ship might be staying close to the suspect ship but they will be standing in parallel. Lastly, after checking for all these factors I select a pair with the lowest speed.

3 results

[Figure 1](#) presents the ship collision 10 minute before and after timeline. We can see that the ships that collided are KARIN HOEJ and MV SCOT CARRIER on December 13th 2021.

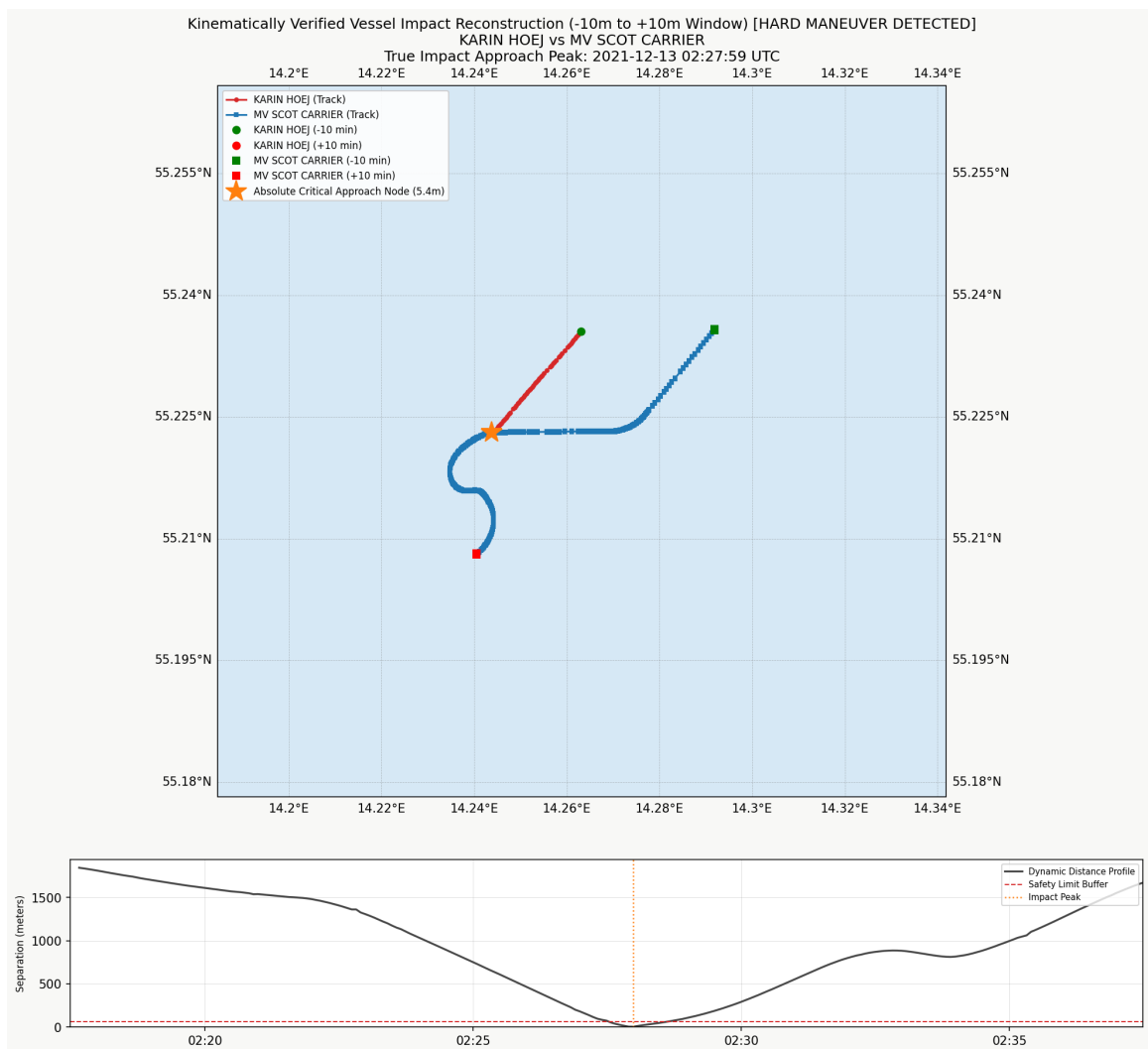


Figure 1: Ship collision timeline